

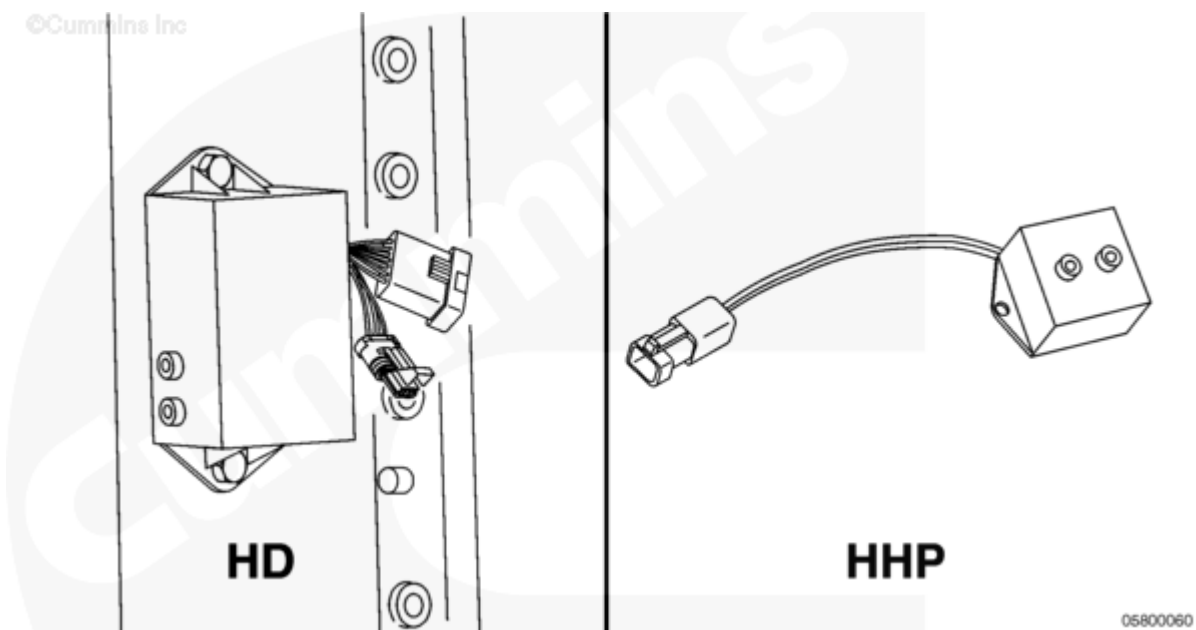
Fault Code: 198

Diagnostic Lamp Fault - Open Circuit

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 198 PID(P): SPN: FMI: Lamp: SRT:	Diagnostic Lamp Fault - Open Circuit. The diagnostic lamp is open.	Centinel™ system status or fault indications will not be properly displayed.



Circuit Description

The Centinel™ control module constantly monitors the operational parameters of the system. The status of the system is displayed by means of diagnostic lamps. There are two Centinel™ diagnostic lamps. They are green and red in color. The green lamp illuminates to indicate that the system's operational parameters are within specified tolerances for heavy-duty **only**. For high-horsepower, the green light indicates the system has power. The red lamp illuminates to indicate that a system parameter is out of specified tolerances.

Component Location

Heavy-duty engines: The diagnostic lamps are located on the Centinel™ control module, located on the make up oil tank.

High-horsepower engines: The diagnostic lamps are located on the diagnostic lamp assembly on the oil make-up tank.

Shoptalk

There is no repair procedure for the heavy-duty. The module **must** be replaced. The following procedure pertains to high-horsepower **only**. Be certain the voltage of the high-horsepower diagnostic lamp assembly matches the application.

Warnings and Cautions

⚠ CAUTION ⚠

To avoid pin and harness damage, use the following test lead when taking measurements: Female AMP, Metri-Pack and Deutsch test lead, Part Number 3822917.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the diagnostic lamps.	
	STEP 1A. Inspect the diagnostic lamps and Centinel™ harness connectors (high-horsepower only.)	No damaged pins
	STEP 1B. Check the diagnostic lamps for an open.	Less than 2M ohms
	STEP 1C. Check the diagnostic lamps for a short.	More than 10 ohms
STEP 2.	Check the Centinel™ harness.	
	STEP 2A. Inspect the Centinel™ harness and the Centinel™ control module connectors.	No damaged pins
	STEP 2B. Check for an open circuit.	Less than 10 ohms

STEPS		SPECIFICATIONS
	STEP 2C. Check for a short circuit to ground.	More than 1k ohms
	STEP 2D. Check for a short circuit from pin to pin.	More than 1k ohms
STEP 3.	Clear the fault codes.	
	STEP 3A. Connect all the components.	Fault Codes 198 inactive

STEP 1. Check the diagnostic lamps.

STEP 1A. Inspect the diagnostic lamps and Centinel™ harness connectors (high-horsepower only.)

Conditions : • Turn the keyswitch OFF. • Disconnect the Centinel™ harness from the diagnostic lamp assembly for high-horsepower systems.		
Action	Specification/Repair	Next Step

• Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Wire insulation damage • Connector shell broken • Damaged locking tab connector. For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	No damaged pins	1B
	Repair the damaged pins Repair or replace the Centinel™ harness or the diagnostic lamp assembly, whichever has damaged pins. • Repair the Centinel™ harness. Refer to Procedure 019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html). • Replace the Centinel™ harness. Refer to Procedure 019-131 (/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html). • Replace the diagnostic lamp assembly. Refer to Procedure 208-001 (/qs3/pubsys2/xml/en/procedures/96/96-208-001.html). • Install the appropriate connector seal if damaged or missing.	3A

STEP 1B. Check the diagnostic lamp assembly for an open.

Conditions :

- Turn the keyswitch OFF.
- Disconnect the diagnostic lamp assembly from the Centinel™ harness.

Action	Specification/Repair	Next Step
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• High-horsepower: Measure the resistance from pin A to pin D on the diagnostic lamp assembly connector for the green lamp circuit and from pin D to pin B for the red lamp circuit. Refer to the wiring diagram for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	Less than 2M ohms	1C
	Replace the diagnostic lamp assembly Refer to Procedure 208-001 (/qs3/pubsys2/xml/en/procedures/96/96-208-001.html).	3A

STEP 1C. Check the diagnostic lamp assembly for a short.

Conditions :

- Turn the keyswitch OFF.
- Disconnect the diagnostic lamp assembly from the Centinel™ harness.

Action	Specification/Repair	Next Step

• High-horsepower: Measure the resistance from pin A to pin D on the diagnostic lamp assembly connector for the green lamp circuit and from pin D to pin B for the red lamp circuit. Refer to the wiring diagram for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 10 ohms	2A
	Replace the diagnostic lamp assembly Refer to Procedure 208-001 (/qs3/pubsys2/xml/en/procedures/96/96-208-001.html).	3A

STEP 2. Check the high-horsepower Centinel™ harness.

STEP 2A. Inspect the Centinel™ harness and Centinel™ control module connectors.

Conditions :

- Turn the keyswitch OFF.
- Disconnect the Centinel™ harness from the Centinel™ control module.

Action	Specification/Repair	Next Step

• Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Wire insulation damage • Connector shell broken • Damaged locking tab connector. For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	No damaged pins	2B
	Repair the damaged pins Repair or replace the Centinel™ harness or the Centinel™ control module, whichever has the damaged pins. • Repair the Centinel™ harness. Refer to Procedure 019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html). • Replace the Centinel™ harness. Refer to Procedure 019-131 (/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html). • Replace the Centinel™ control module. Refer to Procedure 019-130 (/qs3/pubsys2/xml/en/procedures/96/96-019-130-tr.html).	3A

STEP 2B. Check for an open circuit.

Conditions :

- Turn the keyswitch OFF.
- Disconnect the diagnostic lamp assembly from the Centinel™ harness.
- Disconnect the Centinel™ harness from the Centinel™ control module.
- Disconnect the power relay from the Centinel™ harness (heavy-duty applications **only**).

Action	Specification/Repair	Next Step
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• High-horsepower: Measure the resistance from pin 1 of the Centinel™ harness connector to pin A of the diagnostic lamp assembly, harness side.	Less than 10 ohms	2C
• High-horsepower: Measure the resistance from pin 25 of the Centinel™ harness connector to pin B of the diagnostic lamp assembly, harness side. • High-horsepower: Measure the resistance from pin 22 of the Centinel™ harness connector to pin C of the diagnostic lamp assembly, harness side. • High-horsepower: Measure the resistance from pin 10 of the Centinel™ harness connector to pin D of the diagnostic lamp assembly, harness side. Refer to the wiring diagram for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	Replace the Centinel™ wiring harness Refer to Procedure 019-131 (/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html).	3A

STEP 2C. Check for a short circuit to ground.

Conditions :

- Turn the keyswitch OFF.
- Disconnect the diagnostic lamp assembly from the Centinel™ harness.
- Disconnect the Centinel™ harness from the Centinel™ control module.
- Disconnect the power relay from the Centinel™ harness (heavy-duty applications **only**).

Action	Specification/Repair	Next Step
• High-horsepower: Measure the resistance from pin 1 of the Centinel™ harness connector to pin 25. • High-horsepower: Measure the resistance from pin 22 of the Centinel™ harness connector to pin 25. • High-horsepower: Measure the resistance from pin 10 of the Centinel™ harness connector to pin 22. Refer to the wiring diagram for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 1k ohms Replace the Centinel™ wiring harness or Centinel™ control module • Replace the Centinel™ wiring harness. Refer to Procedure 019-131 (/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html). • Replace the Centinel™ control module. Refer to Procedure 019-130 (/qs3/pubsys2/xml/en/procedures/96/96-019-130-tr.html).	2D 3A

STEP 2D. Check for a short circuit from pin to pin.

Conditions :

- Turn the keyswitch OFF.
- Disconnect the Centinel™ harness from the diagnostic lamp assembly.
- Disconnect the Centinel™ harness from the Centinel™ control module.
- Disconnect the power relay from the Centinel™ harness (heavy-duty engines **only**).

Action	Specification/Repair	Next Step
• High-horsepower: Measure the resistance from pins 1, 10, 22, and 25 of the Centinel™ harness connector to all other pins in the connector. Refer to the wiring diagram for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 1k ohms	3A
	Replace the Centinel™ wiring harness or Centinel™ control module • Replace the Centinel™ wiring harness. Refer to Procedure 019-131 (/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html). • Replace the Centinel™ control module. Refer to Procedure 019-130 (/qs3/pubsys2/xml/en/procedures/96/96-019-130-tr.html).	3A

STEP 3. Clear the fault codes.

STEP 3A. Disable the fault code.

Conditions :

- Connect all the components.

Action	Specification/Repair	Next Step

- Start the engine and let it idle for 1 minute. - Verify that Fault Code 198 is inactive.	Fault Code 198 inactive	Complete
	Replace the Centinel™ control module Replace the Centinel™ control module if the fault codes are active. Refer to Procedure 019-130 (/qs3/pubsys2/xml/en/procedures/96/96-019-130-tr.html).	Complete.

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